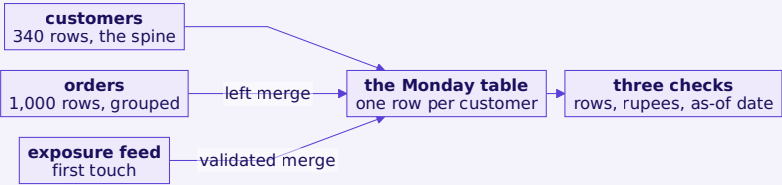


1 A spine, three attachments, three checks



Rows against the customer list (340), spend against Monday's book (Rs 19,84,00,000), and recency from the data's last date (28 September 2026), where the smallest recency is 0.

Crux. groupby is the accumulator automated: split, apply, combine.

2 One row per customer

```
rfm = (orders.groupby("customer_id")
      .agg(last_order=("order_date", "max"),
           frequency=("order_id", "count"),
           monetary=("amount", "sum"))
      .reset_index())
table = customers.merge(rfm, on="customer_id",
                      how="left", validate="one_to_one")

table["frequency"] =
table["frequency"].fillna(0).astype("int64")
AS_OF = orders["order_date"].max()
table["recency_days"] = (AS_OF - table["last_order"]).dt.days
```

Crux. Start from the customer list; groupby only knows the keys it sees.

3 Three tools, one move

The move	SQL	pandas
per customer	GROUP BY	groupby().agg()
attach a source	LEFT JOIN	merge(how="left")
window total	sum() OVER	groupby().transform("sum")
previous month	LAG() OVER	groupby().shift(1)

4 Four defaults that decide a number

Default	What it did	The check
recency from today	win-back 166, truly 111	smallest recency is 0
merge(validate=None)	a repeat copies spend	rows in equal rows out
groupby(dropna=True)	reach 107 at 100 percent	groups add back to rows
pivot_table(aggfunc="mean")	a fall of 18, truly 29 percent	grand total equals source

Crux. Measure recency from the data's last date, never from today.

6 Reshape with the aggregation said

Call	Shape	Answers
pivot_table(index=member, columns=month, aggfunc="sum")	wide	compare along a row
.melt(id_vars=member)	long	follow a trend
pivot(...)	wide, or ValueError on a repeat	one value per cell

Crux. pivot_table averages unless you write aggfunc.

7 Which tool owns which job

SQL owns what Finance audits and reruns; pandas is the analyst's bench for files, merges and five cuts in an afternoon; plain Python shows every step to a reviewer. A hand-edited sheet never computes the source of truth.

Crux. SQL for what Finance audits, pandas for the analyst's bench, plain Python to explain.

THE WORDS ON THIS SHEET

- Grain** What one row of a table is
- Spine** The table every other source is attached to
- Split, apply, combine** What groupby does
- Named aggregation** An agg where each output column is named, with its source and function
- As-of date** The data's last loaded date, carried in the table
- Fan-out** A merge that multiplies rows because a key repeats
- validate=** The merge argument that raises MergeError when keys break a promise
- First touch** One exposure per customer, the earliest